

A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 39

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- Marking points are separated by bullets; each bullet is worth 1 mark unless stated.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	• A waste is assumed to cause pollution until research shows it is unlikely to do so	1
01.2	• If the cost of preventing damage is less than the (penalty for) damage there is a financial incentive to prevent pollution • If polluters can escape their responsibilities / penalties are low, there is little incentive to comply	2
01.3	<i>Any 2 from:</i> • Marine discharges where currents dilute and disperse emissions • Emissions downwind of urban areas • Not discharging waste onto permeable rock above an aquifer	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 243-244; AQA spec 3.4.3.1.	
02.1	• Nuclei of large atoms (uranium-235, plutonium-239) split when hit by a neutron • A small amount of mass is converted to a large amount of energy ($E = mc^2$) and more neutrons are released, causing a chain reaction (controlled by control rods / moderator) • The heat boils water to steam which drives turbines and generators	3
02.2	• Output changes slowly (days) so it cannot follow rapid fluctuations in demand • Very high construction cost but low fuel cost means it is most economic running continuously at full output	2
02.3	• Only 30 t of fuel a year (very high energy density) so no need for railways / ports to deliver millions of tonnes of coal • Still need condenser cooling water (coast / large lake) and a stable site	2
02.4	• Uranium must be mined, purified, concentrated and processed into fuel using (fossil) energy; reactor construction and decommissioning have embodied energy	1

Question	Marking guidance	Marks
02.5	<i>Any 2 from:</i> • No commercial reactor has been fully decommissioned so costs are unknown and often exceed estimates • Large projects often overrun original cost estimates • Long-term waste disposal costs uncertain • State subsidies may hide costs	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 191-193; AQA spec 3.3.4.1.	
03.1	• $(7.7 - 3.9) / 7.7 \times 100$ • = 49.4% (accept 49%)	2
03.2	• Floating sea ice has already displaced its own mass of water; when it melts it contracts and occupies the same volume as the ice below the water line • Melting land ice (glaciers, ice sheets) adds new water to the sea, increasing its volume	2
03.3	• Ice and snow have a high albedo and reflect most sunlight; open sea water has a low albedo • As ice area shrinks more sunlight is absorbed by the sea and converted to heat, raising temperatures • This melts more ice: a positive feedback mechanism; thinner ice melts more easily each summer	3
03.4	• Thermal expansion: warmer sea water expands (slow because of the huge volume and high heat capacity; only surface water warms directly) • Melting of land ice (glaciers, Greenland and Antarctic ice sheets) adds water to the oceans	2
03.5	• Ice shelves grounded on the seabed hold back / block the forward movement of the glaciers or ice sheet behind them; once removed the land ice flows faster (eg West Antarctica)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 94-96; AQA spec 3.2.1.2.	
04.1	• pH 5.6 • Atmospheric carbon dioxide dissolves in rainwater to form (weak) carbonic acid	2
04.2	• Limestone bedrock (calcium carbonate) is alkaline / basic and neutralises the acids as water percolates through it, so the lake pH stays high (7.2) • Granite does not neutralise acids so lake water in A is acidic (4.6) • Low pH mobilises / dissolves aluminium ions from the soil and rock (0.42 mg dm^{-3} in A) which are toxic to fish (damage gills) • Acids also damage fish eggs / gills (exposed living cells) and dissolve the exoskeletons of invertebrate food; lichens and other food organisms lost	4
04.3	• Corrodes metals: railway lines, railings, pipes, pylons, overhead power lines • Dissolves / weakens limestone and marble buildings and statues (eg York Minster)	2
04.4	• Sulfur dioxide is oxidised (by ozone / other oxidants) to sulfur trioxide • Sulfur trioxide dissolves in water droplets to form sulfuric acid (SO₂ alone gives the weaker sulfurous acid)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 250-252; AQA spec 3.4.3.2.2.	

Question	Marking guidance	Marks
05.1	• 25 g ha ⁻¹ (accept 24-27)	1
05.2	• The highest dose that can be applied without killing any members of the population / any individuals of the (non-target) species	1
05.3	<i>Any 3 from:</i> • The MDAF (minimum dose always fatal) of the pest is about 50 g ha⁻¹ • This is higher than the MDNF of the bee (about 20 g ha⁻¹) • So a dose that kills all pests kills some bees / the two curves overlap between about 20 and 50 g ha⁻¹ • A dose of 30-40 g ha⁻¹ would kill most pests with limited bee deaths (compromise)	3
05.4	<i>Any 2 from:</i> • Spray at night / when bees are not flying • Spray when the crop is not in flower • Spray on still days to reduce spray drift • Use a systemic pesticide / seed coating rather than spraying • Use smaller droplets for more even coverage	2
05.5	<i>Any 3 from:</i> • Specificity is the extent to which the pesticide is toxic only to the target species / range of species affected • A highly specific pesticide has low toxicity to non-target species (eg bees, fish, mammals) • So fewer non-target deaths / less loss of predators, pollinators • Fewer indirect (food chain) effects / fewer inter-species relationships lost • Example: pyrethroids have high insect toxicity but low mammal toxicity	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 262-263; AQA spec 3.4.3.2.8.	
06.1	• Trees absorb CO₂ by photosynthesis and store carbon in wood as they grow • Once mature there is no further net storage (photosynthesis balanced by respiration / decomposition), though the standing forest remains a reservoir • Harvesting the timber and keeping it as wood (eg house construction) then replanting produces additional storage as the new trees grow	3
06.2	• Pre-combustion: the fuel is modified before use so no CO₂ is released when burnt - eg gasification of coal to hydrogen and CO₂ (CO₂ stored, hydrogen distributed); oxy-fuel combustion in pure oxygen giving only CO₂ and water vapour which is condensed out • Advantage: capture at a large central facility; suits small dispersed sources (vehicles) that cannot capture CO₂ • Post-combustion: CO₂ removed from exhaust gases by solvents, membrane filtration, adsorption / desorption, cryogenic separation, graphene • Disadvantage: only about 20% of flue gas is CO₂ so separation is costly / energy-intensive; none fully developed	4
06.3	<i>Any 3 from:</i> • Injected into depleted oil / gas fields or saline aquifers • Injected into oil reservoirs to provide pressure for secondary oil recovery • Reaction with fractured basalt to form solid carbonate minerals • Concern: leakage back to the atmosphere / monitoring over centuries; limited suitable geology; cost	3

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 151-152; AQA spec 3.2.4.2.	
07.1	• Spearman's rank correlation coefficient • Mann-Whitney U test • Chi-squared test • t-test	4
07.2	• Chi-squared compares frequencies / counts in categories • Dissolved oxygen is continuously variable measured data, so a t-test or Mann-Whitney U test is needed	2
07.3	• There is always some remaining uncertainty / a probability that the result was produced by chance • Statistics only give a level of confidence (eg 95%, 99%) that the result is significant	2
07.4	• Bars drawn above and below each mean showing plus and minus one standard deviation • If the bars of two means overlap a lot the difference is probably not significant; if they do not overlap it probably is	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 422-423; AQA spec 3.7.2.3.	
08.1	• Molten iron from the blast furnace is converted directly to steel on the same site • So it does not cool and need re-melting; transport energy between sites is also avoided; waste heat from one process used by another (eg Kalundborg)	2
08.2	• Collecting and processing small dispersed quantities can use more energy in transport than is saved • Mass production and large-scale mining have economies of scale; some materials (glass) take more energy to make than plastic but can be refilled many times	2
08.3	• Transmitting at high voltage means low current so less energy is lost as heat in the cables (loss proportional to current squared); overhead cables cheaper than underground which need cooling • ICT / smart grids co-ordinate real-time data on demand and supply from many sources, switching generation, storage (pumped storage, batteries) and flexible loads so surplus is not wasted and inefficient peaking plants are avoided	4
08.4	<i>Any 2 from:</i> • Proximity to the grid / demand to reduce transmission losses and cable costs • Access to cooling water for thermal stations • Resource availability (wind, tidal range, fuel deliveries) • Land availability and planning / public acceptance; stable geology	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 234-235; AQA spec 3.3.4.2.3.	

Question	Marking guidance	Marks
09.1	<i>Any 3 from:</i> • Supply rate of energy crops can be controlled and fuels stored so supply matches demand (not intermittent) • Liquid biofuels (alcohol, biodiesel) can replace petrol / diesel in vehicles • Carbon neutral in principle: CO₂ released equals CO₂ absorbed in photosynthesis • Energy density of alcohol / biodiesel nearly as high as fossil fuels; wastes (straw, sewage, landfill gas) are cheap	3
09.2	<i>Any 3 from:</i> • Large areas of farmland compete with food production or cause clearance of wildlife habitats (eg palm oil) • Intensive farming inputs (fertiliser, pesticide, machinery fuel) may release nearly as much CO₂ as using fossil fuels directly • Low energy density of straw, wood, Miscanthus makes transport uneconomic - must be used locally • Supply from wastes limited to the amount produced	3
09.3	<i>Any 2 from:</i> • Hydrogen from algae: some algae deprived of sulfur produce hydrogen during photosynthesis • Anaerobic digestion extended to crop wastes and marine algae; small-scale digesters (eg one cow's manure supplying a family's cooking gas) • Biofuels from microorganisms (eg oil-producing microalgae)	2
09.4	• Conventional geothermal steam turbines need rocks above 150 °C (recent volcanic activity), too deep to reach economically in the UK • New turbines use fluids such as butane / pentane that boil at low temperatures, so water as cool as 60 °C from deep boreholes / hot dry rock systems can drive them	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 212-214; AQA spec 3.3.4.1.	

Question	Marking guidance	Marks
10.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Context: Lasky's principle - vast quantities exist at low grade; reserves vs resource; COOG affected by price, technology, energy costs • Improved exploration: higher resolution multispectral satellite sensors, IR spectroscopy, gravimetry, magnetometry, seismic, resistivity, portable field analysis, targeted trial drilling • Mechanisation allowing deeper underground mines and larger open-cast mines • Low-energy extraction of low-grade ores: bioleaching with acidophilic bacteria and fungi, phytomining, iron displacement, bacterial adsorption of rare earths, polymer adsorption from seawater • Previously inaccessible deposits: polymetallic nodules, deep sea sediments, Antarctica (currently protected) - environmental and legal issues • Extending use: recycling (advantages, difficulties), Cradle to Cradle design, product lifetime extension, substitution of scarce materials, circular economy • Evaluation: environmental impacts vs benefits; energy costs; economics; which strategies are most sustainable; conclusion	9
10.2	• All of the material that exists in the lithosphere, including that which will never be technologically or economically exploitable	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 130-147; AQA spec 3.2.3.	

Question	Marking guidance	Marks
11.1	Discuss the strategies that can be used to provide sustainable supplies of water for a growing population. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Context: uneven distribution of water, growing population and per capita demand, unsustainable exploitation of rivers and aquifers, hydrological cycle links • New sources: rainwater collection, rivers (discharge, fluctuations, contaminants), reservoirs and site selection, estuary barrages, desalination, unexploited aquifers • Management: artificial aquifer recharge, river-regulation reservoirs, inter-basin transfers, afforestation • Conservation: low-water appliances, xeriscaping, drip irrigation, grey water, leak reduction, metering; pollution control to protect sources • Treatment: matching quality to use; screens, sedimentation, aeration, flocculation, filtration, sterilisation, pH adjustment, activated carbon, reverse osmosis • Evaluation of costs, energy, environmental impacts (habitat loss, methane, inter-tidal habitats, conflicts) and justified conclusion Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 114-127; AQA spec 3.2.2.4-3.2.2.5.	25

Question	Marking guidance	Marks
11.2	Evaluate the contribution of new technologies to the monitoring and understanding of environmental change in the physical environment. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Historical data limitations; proxy data - ice cores, isotope ratios (O-18:O-16, C-14), Mg:Ca in calcite, tree rings, pollen, coral • Satellite sensors: visible light imagery, infrared for vegetation and temperature, radar microwaves for wind, wave height, sea level and oil, weather monitoring; passing through cloud • Sensor carriers: ROVs / drones, AUVs, balloons, aircraft, animals; data recorders for abiotic factors; GPS tracking of floats and sampling sites; Argo floats and ocean monitoring • Climate models and their assumptions; monitoring atmospheric composition (Mauna Loa), ozone (Dobson units), sea level and ice • Applications to hydrosphere, minerals (remote sensing surveys), soils, pollution monitoring • Evaluation: improved coverage and resolution vs cost, calibration, interpretation, remaining uncertainty; justified conclusion Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 9, 89-107, 402-407; AQA spec 3.2, 3.7.	25